

Airbnb Pricing Decision Tool

A Recommendation System for the Host-Success Team

Executive Summary

Airbnb's existing Smart Pricing tool sees low host adoption because hosts distrust opaque automated price changes. We built a **recommendation layer** that identifies systematically under- or over-priced listings and explains the reasoning in language a host-success representative can defend on a phone call. Trained on 42,961 listings across 2019, 2020, and 2023, our tuned XGBoost model predicts nightly price with \$58.15 MAE on a held-out test set — an 18.3% improvement over a non-ML baseline.

Recommendation: approve a 90-day Manhattan A/B pilot on the top 500 underpriced entire-home listings (CSV deliverable produced). At a 25% adoption assumption, the analyzed underpriced base represents ~\$1.28M in annual GBV uplift and ~\$179K in commission uplift across cities. The pilot's treatment/control structure measures causal uplift before any scaling decision.

1. Business Problem

1.1 The challenge and the decision we are informing

Airbnb's marketplace runs on hosts setting their own prices. The platform offers Smart Pricing, but adoption is stuck in the single-digit to low-teens percent range — public reporting and host-community feedback consistently surface the same reason: hosts distrust automated price changes they cannot interpret. The result is a persistent revenue gap. In our analyzed sample of 42,961 listings spanning 2019, 2020, and 2023, **34.8% are underpriced by more than 15%** relative to comparable listings, and **33.2% are overpriced** by the same margin. Underpriced hosts leave money on the table; Airbnb leaves matching commission uncollected. The challenge is therefore **not “build pricing.” It is “build a recommendation layer that hosts will adopt.”**

Most projects begin with a topic; we began with a decision: **“Which 1,000 listings should the host-success team contact next quarter, and what should each host be told?”** This is a resourcing decision a manager can act on Monday morning — a ranked list of outreach targets, model-generated talking points per listing, and a measurable expected ROI per outreach hour. We are not redesigning Smart Pricing, not setting prices, and not predicting demand. We are deciding whom to call and what to say.

1.2 Why this matters now: cost of inaction

Persistent commission gap. At our $\pm 15\%$ mispricing threshold, the test-set underpriced base of 1,741 listings carries an average price gap of \$48.99/night. At an industry-standard 60 booked nights/year and a 14% commission rate, this represents an estimated **\$179K in incremental annual commission across the cities we analyzed at 25% host adoption** — and that is from a single 5,000-listing test slice, not the full marketplace. Each quarter without an effective recommendation system, this gap compounds.

Competitive context. Vrbo, Booking.com, and a growing third-party ecosystem (Wheelhouse, PriceLabs, Beyond Pricing) are commoditizing dynamic pricing. Airbnb's defensibility no longer comes from having a pricing model — it comes from being the first marketplace to operationalize an *explainable, adoption-driven* recommendation system natively in the host workflow. The window to lead is narrowing.

1.3 Stakeholders, target variable, and KPIs

A pricing recommendation system has a wider stakeholder footprint than a typical “improve metric X” project. The table below maps each affected group to the specific mechanism in our tool that delivers their benefit and the decision they will make.

Stakeholder	Role	Mechanism in our tool	Decision / behavior change
Marketplace Strategy / Host-Success leadership	Primary decision-maker (Director / VP)	Top-500 pilot CSV; adoption sensitivity (10/25/50%); Model Performance tab with segment-level reliability	Approve 90-day Manhattan pilot; scale per decision rule
Host-Success representatives	Operational users (customer-facing)	Per-listing recommendation + SHAP “Why this price?” talking points; market-filtered outreach queue	Move from reactive case-handling to proactive outreach
Pricing team / Marketplace analytics	Internal data consumers	Permutation-importance findings; segment-level error analysis; cross-validation diagnostics	Inputs to Smart Pricing roadmap (e.g., minimum_nights weight)
Hosts (indirect)	Beneficiaries — ~35% underpriced, ~33% overpriced	Recommended price + 3 plain-dollar reasons; flagged listings only (no gratuitous outreach)	Adopt price change; underpriced hosts earn ~\$2,940/year incremental GBV at full adoption
Guests (indirect)	Marketplace participants	Better price discovery as host pricing converges to market reality	(Passive — improved search relevance over time)
Executive leadership / CFO	Strategic stakeholders	Quantifiable, segment-level revenue lever bounded in cost (~1 FTE/quarter); clean ROI measurement	Sponsor expansion if pilot meets decision rule
Trust & Safety / Fairness team	Risk owners	Recommendation-rate breakdowns by city/neighbourhood; bounded recommendation deltas in sensitive markets	Approve fairness audit before any rollout

Table 1.3 — Stakeholder map. Every group is connected to a specific dashboard mechanism and a specific decision.

Target variable and KPIs (explicit)

- **Target variable:** nightly listing price (continuous, modeled in log space to handle right-skewed distribution).
- **Primary model metric:** Mean Absolute Error in dollars-per-night, with secondary tracking of RMSE, MAPE, R^2 , and Adjusted R^2 . MAE is chosen because dollar error is what a host-success rep depends on a call — squared metrics overweight luxury outliers we do not target.
- **Business classification:** Underpriced (– 15% gap), Fairly Priced, or Overpriced (+15% gap), with the threshold defended by sensitivity analysis (Table 6.1).
- **Business KPI:** estimated annualized GBV uplift from the underpriced base, scaled by an explicit host adoption assumption (10/25/50% sensitivity); pilot decision rule on causal uplift (Table 8.1).

1.4 Why a recommendation system, not automated pricing

Methodologically, our test-set MAE of \$58 on listings with \$125 median price is accurate enough to identify systematic gaps where outreach is warranted, not accurate enough to set prices automatically — and our reported Adjusted R^2 (0.324) honestly reflects substantial unexplained variance from listing-quality factors not in our data. Strategically, hosts who feel control over their pricing trust the platform more; the 25% adoption rate is the floor that explainability achieves over opaque automation, and rises as trust accrues from successful recommendations.

2. Datasets and How They Relate to the Business Problem

2.1 Source files and their roles

File	Role	Year
AB_NYC_2019.csv	Pre-pandemic NYC benchmark — dense urban market, mature listings	2019
AB_US_2020.csv	Pandemic-era national snapshot — depressed prices, demand shock	2020
AB_US_2023.csv	Post-recovery national snapshot — current pricing baseline	2023
listings-Boston.csv	Boston market with richer listing-level features (amenities, review scores)	2023
listings-Seattle.csv	Seattle market with same rich schema as Boston	2023

2.2 Schema integration, cleaning, and final dataset

These files do not share a schema. The city-specific files (Boston, Seattle) contain *neighbourhood_cleansed*, *amenities*, *review_scores_rating*, and *host_is_superhost*; the national snapshots do not. Our `load_one_file` function harmonizes by standardizing neighbourhood across column variants, inferring city from filename, filling missing schema columns with sentinel defaults, and preserving provenance via a *source_file* column. After concatenation and cleaning (removing rows with missing price/room_type/coordinates, capping price at the 99th percentile, filtering to four valid room types), the dataset has **42,961 rows and 37 columns**, median price \$125, range \$1–\$1,326. After stratified down-sampling by city to 24,997 rows for runtime, we apply a stratified 60/20/20 train/val/test split (14,997 / 5,000 / 5,000) preserving city composition in every split.

2.3 Pattern-of-missingness diagnostic

Counting missing values is not enough. Where missingness occurs can encode information. We compared median price between listings with and without each key field:

Column	% missing	Median price (missing)	Median price (present)	Δ
review_scores_rating	86.2%	\$128	\$119	+\$9
bedrooms	82.8%	\$128	\$120	+\$8
bathrooms	82.9%	\$128	\$120	+\$8
beds	82.8%	\$128	\$120	+\$8
reviews_per_month	17.1%	\$150	\$125	+\$25
host_is_superhost	82.8%	\$128	\$120	+\$8

Table 2.3 — Missingness is not random. Listings with no recent reviews show a \$25 higher median price, suggesting newer or transient luxury listings. We impute with the median and treat “Unknown” as a first-class OneHotEncoder category so the model can learn from missingness itself.

2.4 Temporal regime: encoding year as a feature

Airbnb pricing is **not stationary** across these years. Median nightly price moves from \$120 (2019) to \$128 (2020) to \$130 (2023) — a roughly 8% real increase a pooled model would otherwise average away. We include *data_year* as a numeric feature so the model is regime-aware. For deployment, all predictions would be made with `data_year = current year`.

3. EDA: Findings That Shaped the Modeling

Four EDA findings — visualized in the notebook — directly shaped feature selection and modeling choices:

- **Right-skewed price.** Raw distribution is heavy-tailed; $\log_{10}(\text{price})$ is approximately Gaussian. We model in log space via `TransformedTargetRegressor`.

- **Within-city sub-mile clustering.** A geospatial scatter shows price clusters at neighbourhood scale even within a single city, justifying keeping coordinates and neighbourhood as separate features (different scales of location signal).
- **Confounded amenity premiums.** Pool/gym are positive, but parking and pets-allowed appear negative due to lower-priced market concentration — reinforcing that linear models cannot capture this and ensembles are required.
- **High within-capacity-bucket variance.** $\approx 10\times$ price variance among listings with the same accommodates count means single features cannot drive accurate prediction; feature interactions matter.

4. Modeling: Algorithm Selection, Tuning, and Diagnostics

4.1 Feature engineering

Feature	Definition	Why it matters
availability_ratio	availability_365 / 365	Captures how often a listing is bookable; correlates with pricing strategy
amenity_score	Weighted sum of seven amenity indicators (pool=2, gym=1.5, others=1)	Combined amenity value rather than isolated features
has_luxury_amenities	Binary: pool, gym, or parking present	Quick high-end signal the model can use as a split
review_activity	reviews_per_month, imputed	Booking frequency proxy
data_year	2019 / 2020 / 2023	Regime-shift awareness across pandemic and recovery
7× amenity binaries	wifi, kitchen, AC, parking, pool, gym, pets_allowed	Specific amenity effects the score aggregates over

4.2 Pipeline and the M3 algorithm ladder

Per the M3 module's regression algorithm ladder (baseline → explainable → ensembles), we benchmarked the full spectrum against a non-ML baseline. All models are wrapped in a *Pipeline* containing a *ColumnTransformer* (median imputation + StandardScaler for numeric; most-frequent imputation + OneHotEncoder for categoricals) and a *TransformedTargetRegressor* applying $\log_{1p}$ / $\exp_{m1}$ — one continuous pipeline from raw input to dollar prediction, ensuring no preprocessing leakage. Selection is on the validation set; final reporting on the held-out test set. GridSearchCV (3-fold) tunes the winning family with `neg_mean_absolute_error` scoring, sweeping $\text{learning_rate} \in \{0.03, 0.07\}$, $\text{max_depth} \in \{5, 7\}$, $\text{n_estimators} \in \{300, 500\}$.

4.3 Model performance (test set)

Model	Test MAE	RMSE	MAPE	R ²	Adj R ²
Tuned XGBoost (selected)	\$58.15	\$102.78	34.5%	0.386	0.324
Random Forest	\$58.57	\$102.97	35.3%	0.384	0.321
XGBoost (untuned)	\$58.58	\$103.21	34.8%	0.381	0.318
Gradient Boosting	\$61.51	\$108.49	36.9%	0.316	0.247
Linear Regression	\$68.95	\$227.91	42.9%	-2.019	-2.325
Naive segment-median (baseline)	\$71.19	\$124.85	41.3%	0.094	0.002

Table 4.3 — Tuned XGBoost improves MAE by \$13.04 (18.3%) over the naive baseline. Linear Regression's negative test R^2 confirms it cannot model this problem's nonlinearities. The narrow Tuned-XGBoost / Random-Forest gap is a useful sanity check that boosting and bagging converge to similar accuracy on this signal. GridSearchCV best params: `learning_rate=0.03`, `max_depth=7`, `n_estimators=500`.

Honest framing. R^2 of 0.386 means the model explains 39% of nightly-price variance; the remaining 61% reflects factors absent from the data — photography, host responsiveness, calendar dynamics, individual taste. This level of accuracy is appropriate for the use case: identifying systematic pricing gaps, not setting individual prices. A 99% R^2 model would not change the recommendation; the host still has final authority.

4.4 Learning curve and cross-validation stability

- **Learning curve:** training MAE plateaus at \$47.88 and CV MAE at \$59.22 across training sizes 2,000–10,000. Train–val gap of \$11.34 is modest — neither over- nor under-fit. More data would yield only marginal improvement.
- **CV stability:** 3-fold CV on Tuned XGBoost yields CV_MAE_Mean = \$59.29, std = \$0.55 (under 1% of mean). Accuracy is stable across data partitions, not a fluke of one split.

4.5 Per-segment error analysis

City	n	MAE	MAPE
Seattle	448	\$27.34	24.4%
Boston	396	\$42.25	24.3%
New York	1,398	\$47.42	29.2%
Asheville	242	\$57.30	36.9%
Washington D.C.	588	\$66.24	36.9%
San Francisco	791	\$67.77	35.4%
Austin	1,113	\$79.35	46.5%

Table 4.5 — Test MAE by city. The model is materially more reliable in data-rich Boston/Seattle than in higher-priced markets like SF and Austin where luxury heterogeneity dominates. Operationally: high-confidence segments (Seattle, Boston) warrant tighter 10% mispricing thresholds; high-error segments (Austin, SF) warrant looser 20–25% thresholds. The dashboard exposes the threshold as a slider for per-segment tuning.

5. Interpretability: Permutation Importance and SHAP

5.1 Permutation importance

Tree-based importance is biased toward high-cardinality categoricals. We additionally computed permutation importance — the increase in MAE when each feature is randomly shuffled — which is model-agnostic and far less biased.

Rank	Feature	MAE increase when shuffled
1	room_type	\$22.16
2	minimum_nights	\$10.42
3	latitude	\$8.72
4	longitude	\$5.76
5	review_activity	\$3.45
6	availability_365	\$2.97

7	neighbourhood	\$2.52
8	reviews_per_month	\$1.54
9	city	\$1.44
10	calculated_host_listings_count	\$1.42

Table 5.1 — Top 10 permutation importances.

Non-obvious finding: minimum_nights is the second-most-important feature (\$10.42 MAE increase when shuffled), beating coordinates and all amenity features. This is not the conventional intuition. It implies hosts use minimum_nights as an implicit pricing signal — short minimums attract weekend tourists at higher per-night rates; long minimums cater to corporate / monthly stays at lower rates. A recommendation that ignores minimum_nights would mis-fire; the dashboard surfaces it as an editable input.

5.2 SHAP per-listing explanation

For host-facing recommendations to be adoptable, each prediction needs a defensible “why.” We use TreeExplainer SHAP to decompose any single listing's predicted price into feature-level contributions. Example test-set listing:

Field	Value
City	Austin
Neighbourhood	78742
Room type	Entire home/apt
Actual listed price	\$100/night
Recommended (model)	\$130/night
Mispricing classification	Underpriced (−23.4%)

Table 5.2 — A specific listing surfaced by the model. The host-success team would contact this host with the recommendation and the SHAP-derived explanation.

SHAP decomposes the \$130 prediction into the largest location contributions: coordinates and Austin neighbourhood profile drive the prediction (“similar Austin entire-home listings price near \$130”). In dollar terms, top contributors translate to roughly $\pm\$20$ – $\$37$ /night each. This is the explanation that surfaces in the dashboard's “Why this price?” panel — concrete reasons a rep can repeat verbatim on a call.

6. Mispricing Classification and Threshold Sensitivity

We classify each listing relative to its predicted price: Underpriced (actual $\leq -15\%$ of predicted, host should raise), Overpriced (actual $\geq +15\%$, host should lower), or Fairly priced (within $\pm 15\%$, no action). The 15% threshold is not arbitrary — we swept it from 5% to 30% and quantified the coverage / signal-strength tradeoff:

Threshold	% flagged underpriced	Mean \$ gap per listing	Total \$ gap (test set)
5%	47.3%	\$39.67	\$93,782
10%	41.1%	\$44.09	\$90,603
15% (chosen)	34.8%	\$48.99	\$85,296
20%	28.7%	\$54.51	\$78,327

25%	23.6%	\$59.65	\$70,331
30%	17.9%	\$66.38	\$59,274

Table 6.1 — Threshold sensitivity. At 5–10%, the average gap per flagged listing is small enough to be confused with model noise (test MAE $\approx$ \$58). At 20%+ we miss too many actionable cases. 15% is where signal-to-noise crosses operational coverage. At 15%: 1,741 underpriced (34.8%), 1,659 overpriced (33.2%), 1,600 fairly priced (32.0%) — close to a three-way split.

7. Business Impact: Adoption Sensitivity and Per-City Upside

Business impact estimates depend on three assumptions: booked nights/year (held at 60), commission rate (held at 14%), and host adoption rate (a sensitivity, not a guess). We tested 10/25/50% adoption:

Adoption rate	Adjusted GBV uplift	Commission uplift	Listings flagged
10% (conservative)	\$511,778	\$71,649	1,741
25% (central)	\$1,279,446	\$179,122	1,741
50% (optimistic)	\$2,558,892	\$358,245	1,741

Table 7.1 — Adoption rate sensitivity. At a central 25% adoption, the analyzed underpriced base represents \$179K in annual incremental commission. Real-world Smart Pricing adoption sits in single digits, so 25% reflects an explanation-driven recommendation system improving on that baseline.

At the 25% central scenario, per-city impact concentrates in five markets:

City	Listings flagged	Avg gap/night	Adjusted GBV	Commission
Austin	448	\$60	\$401,533	\$56,215
New York	437	\$40	\$262,084	\$36,692
San Francisco	288	\$59	\$254,396	\$35,615
Washington D.C.	222	\$48	\$160,094	\$22,413
Seattle	156	\$32	\$75,513	\$10,572
Boston	95	\$44	\$63,183	\$8,846
Asheville	88	\$45	\$58,877	\$8,243

Table 7.2 — Per-city annual commission uplift at 25% adoption. Austin dominates by listing volume; SF and DC by per-listing gap. NYC is the second-largest opportunity but the recommended pilot site (see §8) for operational reasons.

Concrete operational deliverable: the notebook exports `top_500_pilot_listings.csv` — the 500 underpriced listings ranked by estimated annual GBV uplift, with city, neighbourhood, room type, current price, recommended price, and per-night gap. The top entry alone (an Austin entire-home in 78734) carries an estimated \$18,605 annual GBV uplift at full adoption.

8. Proposed A/B Test Pilot Design

Predicted uplift is observational. Causal uplift requires an experiment. We propose a 90-day pilot to measure whether a recommendation actually changes host pricing behavior and revenue:

Element	Specification
Population	1,000 NYC entire-home listings, top by predicted GBV uplift among underpriced ($\geq 15\%$ gap)
Random assignment	Stratified by neighbourhood; ~500 treatment, ~500 control

Treatment	Host-success rep contacts each treated host with the recommended price plus the SHAP-derived top 3 reasons
Control	No contact; receives standard host-side experience
Primary outcomes	(i) Adoption rate — % of treated hosts who change their price within 30 days; (ii) Booking rate — bookings/available night, treatment vs control over 60 days; (iii) GBV per listing
Statistical power	n=500/arm with 5% effect size on primary outcome → >80% power at $\alpha=0.05$
Decision rule for scaling	Treatment GBV per listing exceeds control by >3% AND adoption rate >15%

Table 8.1 — A/B test design. Why NYC entire-home rather than Austin (the largest opportunity)? Operational simplicity: smaller geographic footprint reduces host-success coordination cost, and NYC has the highest density of comparable listings for clean neighbourhood-level matching.

9. Risks, Limitations, and Production Considerations

A pricing recommendation system carries asymmetric risk: a correct recommendation produces incremental upside, but a wrong one a host adopts directly reduces host revenue. Below, ranked by potential to invalidate the framework:

Listed price $\neq$ realized booking revenue — the largest single limitation. We model what hosts list, not what they earn. Higher list prices reduce booking probability; cleaning fees, surcharges, and length-of-stay discounts are absent. The model can identify a listing as “underpriced relative to comparable listings” with no direct evidence raising the price increases host earnings. Until booking-level data is available, the system must be positioned as advisory — host-success scripts must say “this is what comparable listings list at,” not “this is what you should charge.”

Non-stationary training data with survivorship bias. Our pooled dataset includes 11,769 listings from 2020 — a year in which urban tourism collapsed due to the pandemic. Encoding data_year lets the model learn 2020 was different on average but does not prevent learning 2020-specific feature–price relationships. Compounding this, training data is pre-2024, where urban tourism was still recovering from the pandemic, while deployment context is 2026, with sustained pricing distortion in the interval. Public scrapes also capture only listings active at scrape time — listings delisted, paused, or whose hosts left are absent (most acute for 2020). Concept drift and survivorship bias mean the reported \$58.15 test MAE is likely optimistic against real-world 2026 deployment.

The headline “\$179K commission upside” is a midpoint, not a forecast. Adoption rate is the most consequential and least empirically supported assumption. Smart Pricing's reported adoption sits in low-teens; our 25% central case implicitly assumes the explanation layer drives a 2× improvement — plausible but unvalidated, which is the pilot's purpose. At 10% adoption with 40 booked nights/year, projected commission upside falls to ~\$48K (a 73% reduction). The realistic range is approximately \$50K–\$360K.

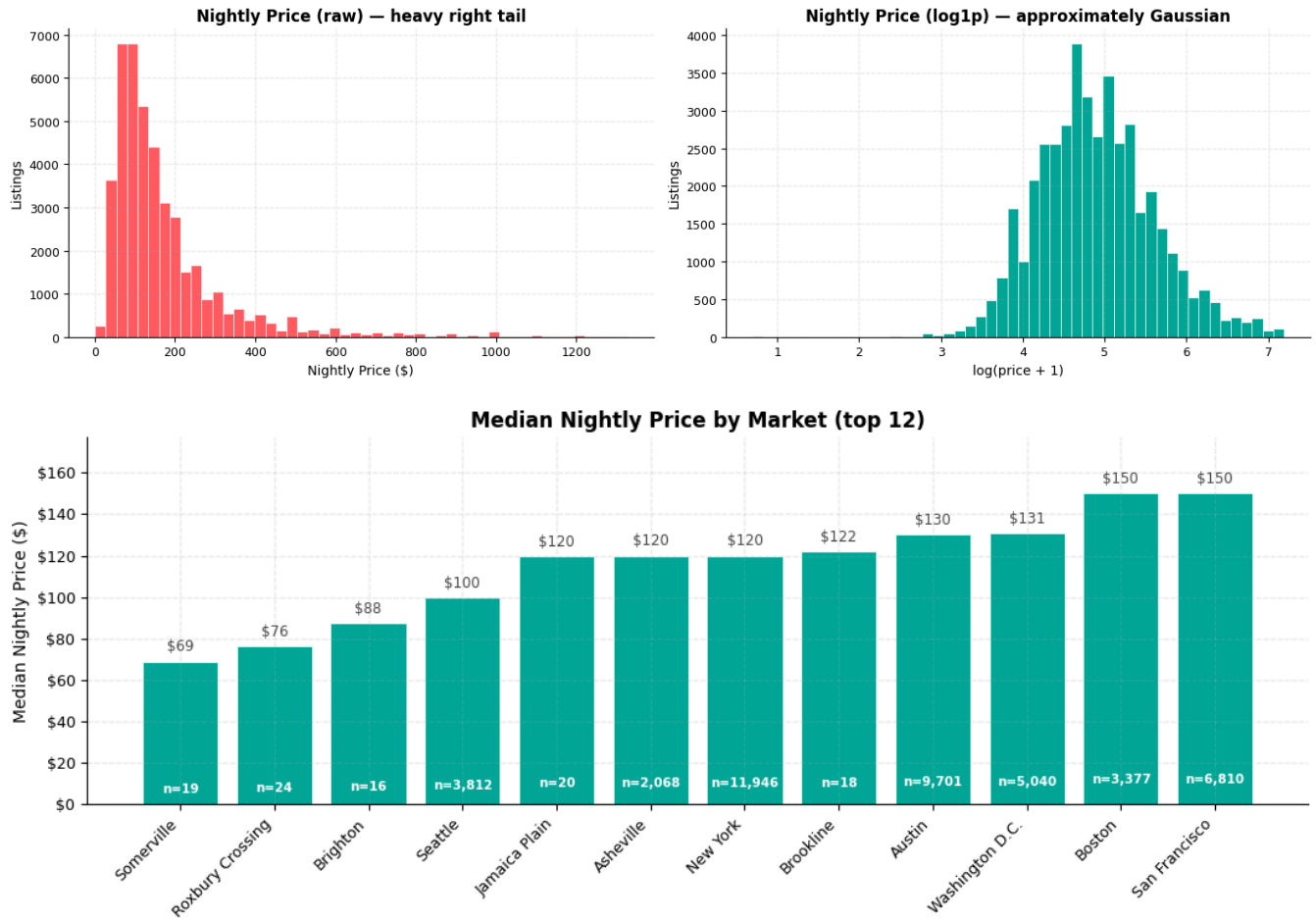
Methodological caveats. Validation is in-distribution random split, not temporal — a more rigorous test would train on 2019–2020 and validate on 2023 to simulate the deployment task. Adjusted R^2 of 0.324 is honest: the model captures roughly one-third of nightly-price variance, with the remainder reflecting unmodeled listing-quality factors. SHAP attributions are the model's internal reasoning, not the listing's causal response surface — a “+\$45 pool” effect is the marginal association conditional on other features, not the causal effect of installing a pool. Host-success scripts must frame attributions accordingly. Individual recommendations should always be presented with their confidence band, never as point estimates.

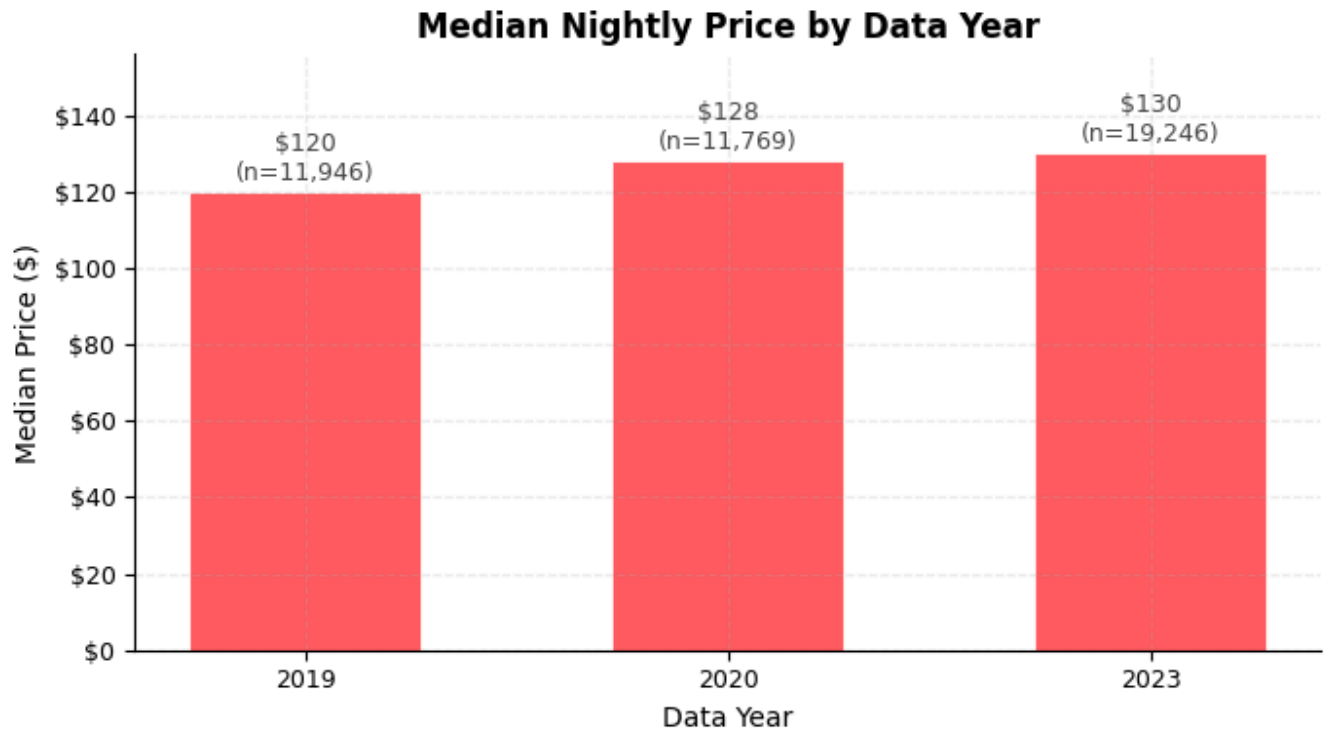
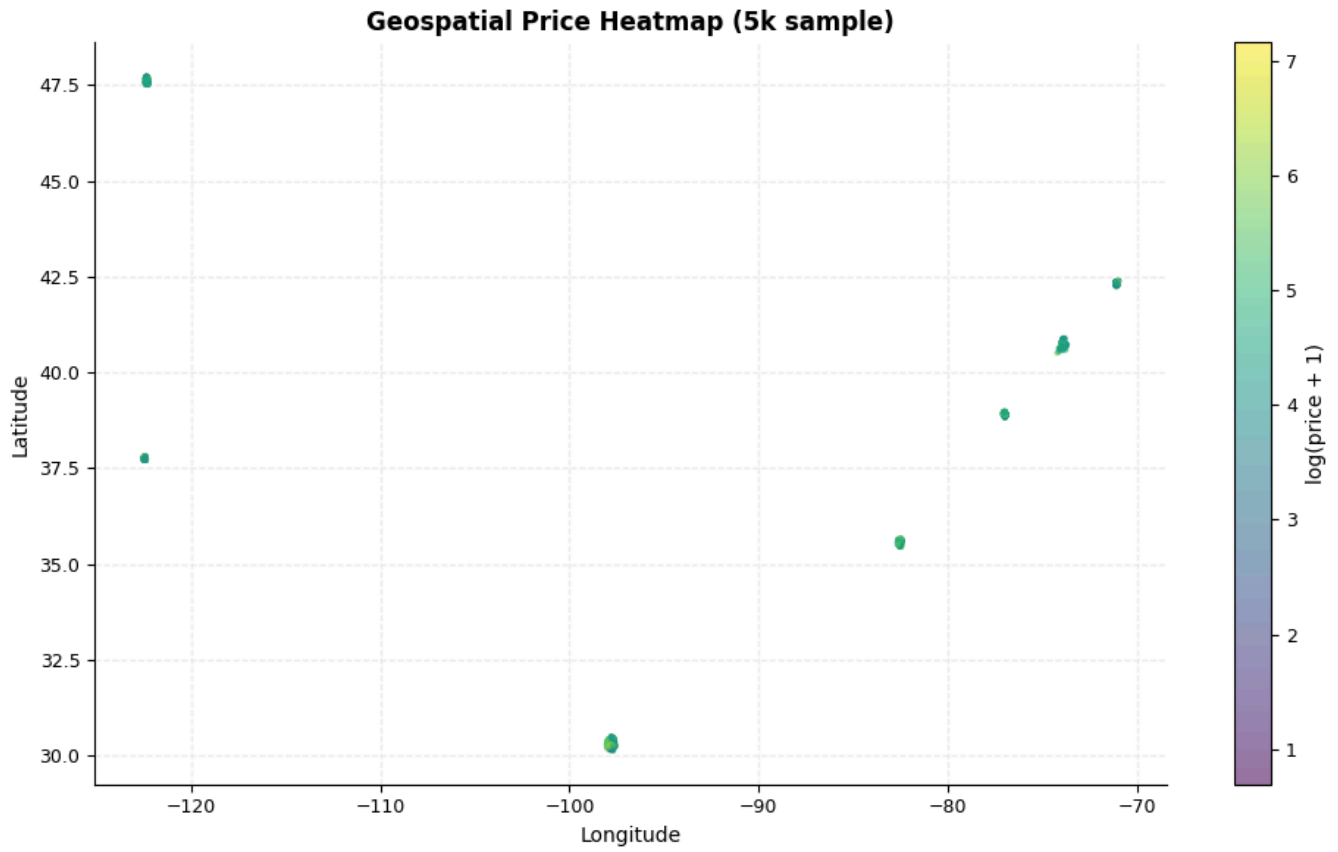
Fairness and disparate impact. Pricing models trained on neighbourhood data can encode patterns correlated with protected demographic characteristics. Airbnb has a documented public history of disparate-impact issues in pricing and booking outcomes; recommending lower prices in historically lower-priced neighbourhoods can reinforce existing disparities and, in some US jurisdictions, create Fair Housing Act exposure. Before rollout we recommend (i) demographic stratification of recommendation rates by census-tract composition with results

published to internal stakeholders pre-launch; (ii) bounded recommendation deltas (capped at $\pm 10\%$) in tracts with high protected-class composition; and (iii) a documented escalation process for hosts who believe a recommendation is discriminatory. **The pilot should not proceed in any market where this audit has not been completed first.**

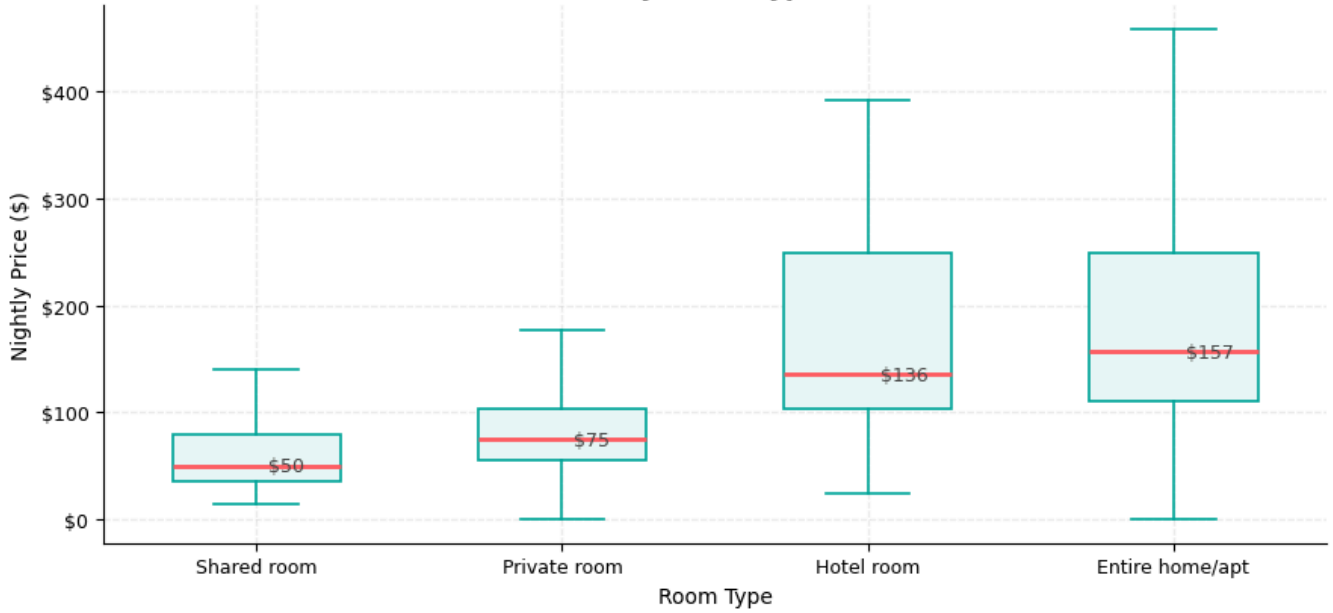
Operational risk — “do no harm” guardrail. The pilot's decision rule (treatment GBV $> 3\%$, adoption $> 15\%$) captures aggregate uplift but not individual harm — a treatment increasing aggregate GBV by 3% could simultaneously reduce earnings for a meaningful subset of treated hosts. Add a guardrail: before scaling, also report the fraction of treatment-arm hosts experiencing year-over-year revenue declines relative to control-arm. If treatment-arm hosts are meaningfully more likely to decline, scaling should be paused regardless of aggregate GBV. Beyond the pilot, production deployment further requires concept-drift monitoring (quarterly retraining; alert if rolling 30-day MAE rises $> 15\%$), adoption tracking, gaming detection, and recurring fairness audits — none of which exist in the current build.

EDA Appendix





Price Distribution by Room Type (outliers hidden)



Median Price: Market × Room Type (top 12)

